Luís Vilaça

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https://lvilaca16.github.io/

About me

B.Sc. and M.Sc. in Electrical and Computer Engineering, and currently attending the Ph.D. at FEUP. He is also an invited teaching assistant at ISEP. Since early in his academic path, he developed a particular interest in Computer Vision (CV) and Machine Learning (ML). Although these topics were outside the curricula, Luís challenged himself with projects related to Facial Detection and Recognition. He scored 18 and 20 (over 20) in his B.Sc. and M.Sc. dissertation, respectively.

The experience obtained by working as a research assistant at INESC TEC on several national and international projects allowed him to grow as a goal-oriented person, and to develop skills to adapt to new and challenging scenarios.

During this period, he was involved in three different Multimedia-related projects, exploring emergent concepts and tools of CV and ML for various applications. He also organized workshops and was involved in the co-supervision of master's thesis, summer internship mentoring, and presentations of ongoing work for dissemination purposes or in project meetings. Besides the scientific aspects, participating in these activities also enabled him to develop further his soft skills, from which he started to understand how to redirect information to reach different audiences.

His research interests include computer vision, multimodal machine learning, and multimedia recommendation. Luís is actively seeking new and challenging professional and academic experiences in CV and ML.

Education

2021-(Curr) PhD in Electrical & Computer Engineering
Faculty of Engineering, University of Porto (FEUP)

2018-2021 M.Sc. in Electrical & Computer Engineering
School of Engineering, Polytechnic Institute of Porto (ISEP)
Classification: 17 (over 20)
Development of my MSc Dissertation, entitled "Towards a Scalable Dataset Construction for Facial: A guided data selection approach for diversity stimulation", with a final grade of 20 (over 20).

2013-2018 B.Sc. in Electrical & Computer Engineering
School of Engineering, Polytechnic Institute of Porto (ISEP)
Classification: 14 (over 20)
Development of my BSc Dissertation, entitled "Facial Detection and Recognition", with a final grade of 18 (over 20).

Experience

10/18-(Curr.) Research Assistant — Porto, Portugal
at INESC TEC

09/22-(Curr.) Invited Assistant Professor — Porto, Portugal
at FEUP - Faculty of Engineering, University of Porto
I instructed practical classes in "Information Processing and Retrieval" to first-year master students.

01/24-08/24 Research Intern & Visiting Researcher — Tokyo, Japan
at NII - National Institute of Informatics
Implementing algorithms for audiovisual alignment using natural language. Collected and distributed a dataset for the State-of-the-Art for Audio-Video-Language learning. Implemented and compared strategies for evaluating Large Language Models. All implementations were done in Pytorch.

03/22-02/24 Invited Assistant Professor — Porto, Portugal
at ISEP - School of Engineering, Polytechnic Institute of Porto
I instructed practical classes in "Data Communications," "Multimedia Communications/Multimedia Systems," and "Advanced Computational Systems" to second and third-year undergraduate students.

01/21-01/24 Developer @ CTM Slurm Dev. Team — Porto, Portugal
at INESC TEC
Developing and maintaining the computational structure of CTM for its researchers (DevOps/MLOps). I was envolved in the development of the computational infrastructure for ML-based experiments within CTM. The infrastructure was based on Slurm. I was in contact with HPC systems, virtualization technologies, e.g. Docker, VirtualBox, Hyper-V, VMWare, and IaC/S software, e.g. Ansible and OpenStack. The project was entirely automated using Gitlab CI/CD mechanisms for generating new Docker images and pushing changes into production automatically.

10/21-03/22 Research Intern — (online) Tokyo, Japan
at NII - National Institute of Informatics
Implemented algorithms for audiovisual representation learning and cross-modal retrieval.

Skills

C++, HTML, JavaScript & PHP

SQL (MySQL, PostgreSQL)

Ansible

Docker

C, Assembly

Python

Tensorflow, Keras⋆3
Scikit-learn, Numpy⋆4
Matlab⋆4
Matplotlib, Seaborn⋆4
Pytorch⋆4
Computer Architectures⋆4
Computer Networks⋆4

(*)[Scale: 0 (Fundamental Awareness) to 6 (Expert).]

Research Projects

10/18-06/19 TEC4Growth - RL FourEyes - Intelligence, Interaction, Immersion and Innovation for media industries, Proj. ID: NORTE-01-0145-FEDER-000020
at INESC TEC

10/18-06/19 CHIC: Cooperative Holistic view on Internet and Content, Proj. ID: POCI-01-0247-FEDER-024498
at INESC TEC
Implementing algorithms for Facial Detection and Recognition for multiple personalities of interest using data-driven approaches in Tensorflow/Keras. While, at the same time, studying ways to reduce its resource use, namely images and time spent in training.

10/18-06/19 FotoInMotion: Repurposing and enriching images for immersive video storytelling, Proj. ID: H2020-ICT-20-2017-RIA-780612
at INESC TEC
Implementing algorithms for Object Detection mainly using the R-CNN framework. The goal was to describe a given picture through the visual content information, namely objects, by using Deep Learning algorithms. Subsequently, integrating them in the FiM system and deploying solutions for the users.

Publications

Jour. Paper L. Vilaça, Y. Yu, and P. Viana. "A Survey of Recent Advances and Challenges in Deep Audio-Visual Correlation Learning" in ACM Computing Surveys, 2023. (In review)
arxiv link (old version): https://arxiv.org/abs/2202.13673

Jour. Paper L. Vilaça, P. Viana, P. Carvalho, and M.T. Andrade. "Improving Efficiency in Facial Recognition Tasks through a Dataset Optimization Approach" in IEEE Access, 2024.

Conf. Paper L. Vilaça, P. Viana, P. Carvalho, and T. Andrade. "Improving Audiovisual Content Annotation Through a Semiautomated Process Based on Deep Learning" in International Conference on Soft Computing and Pattern Recognition, pp. 66–75, Springer, 2018.
doi: 10.1007/978-3-030-17065-3_7

Conf. Paper P. Viana, P. Carvalho, M. T. Andrade, P. P. Jonker, V. Papanikolaou, I. N. Teixeira, L. Vilaça, J. P. Pinto, and T. Costa. "Semantic storytelling automation: A context-aware and metadata-driven approach" in Proceedings of the 28th ACM International Conference on Multimedia (Core A*), pp. 4491–4493, 2020.
doi: 10.1145/3394171.3416528

Conf. Paper P. Viana, P. Carvalho, M. T. Andrade, I. N. Teixeira, P. P. Jonker, L. Vilaça, J. P.Pinto, and T. Costa, "From a still image to a semantically aware video: A context and metadata-driven automatic media production framework", in the 17th ACM SIGGRAPH European Conference on Visual Media Production, 2020.

Conf. Paper Teixeira, Inês N.; Viana, P.; Andrade, Maria Teresa; Carvalho, Pedro; Vilaça, Luis; Pinto, José Pedro; Costa, Tiago; Rapanakis, Stamatis; Jonker, Pieter P., "Creating automatic storytelling videos from still images: a semantically-aware approach", in ICPR 2020: International Conference on Pattern Recognition 2020 (Core B), 2020.

Curricular Projects

09/2020 **Towards a Scalable Dataset Construction for Facial, A guided data selection approach for diversity stimulation:** The work developed in this dissertation aims at investigating the impact of selecting a reduced amount of images in a Facial Recognition, when using Deep Learning. The solution proposed uses a feature engineering pipeline to discriminate the diversity of faces by increasing the amount of information. The goal is to maintain the algorithm's throughput while reducing the amount of images used.

09/2018 **Facial Detection and Recognition:** Pipeline to generate and validate image annotations, it was created to improve the audiovisual content annotation through a semi-automated process based on Deep Learning. This process can also accelerate the training phase of a DL/Machine Learning model.

06/2018 **Pre-processing in computer vision:** Research project. The importance of pre-processing in Computer Vision applications. It were also created learning guides on Computer Vision (OpenCV) and Machine Learning (Scikit-Learn).

01/2018 **High Speed Autonomous Guided Vehicle:** High speed line follower with a PID control.

Activities

09/19-06/20 Co-supervision of a Master Thesis: "Automatic Emotion Identification: Facial Expressions Analysis and Detection in films", by João Carlos Miranda de Almeida
at INESC TEC - Institute for Systems and Computer Engineering, Technology and Science, Portugal

05/19 Machine/Deep Learning Workshop - Openday CTM (2019/05/15 - 2019/05/17)
at INESC TEC
Workshop (Member of the Organising Committee)

07/19-08/19 Mentoring of Summer Internships - Influence of the Diversity of Faces
at INESC TEC
Student name: João Varela and Bruno Costa

07/21 Participant at LxMLS - Lisbon Machine Learning Summer School 2021

07-08/22 Participant at OxML - Oxford Machine Learning Summer School 2022

07-08/24 Participant at OxML - Oxford Machine Learning Summer School 2024

01/21-(Curr) Developer @ CTM Slurm Dev. Team - Developing and maintaining the computational structure of CTM for its researchers (DevOps/MLOps)
at INESC TEC
During this time I developed from root the computational infrastructure for ML-based experiments within CTM. The infrastructure was based on Slurm. I was in contact with HPC systems, virtualization technologies, e.g. Docker, VirtualBox, Hyper-V, VMWare, and IaC/S software, e.g. Ansible and OpenStack. The project was entirely automated using Gitlab CI/CD mechanisms for generating new Docker images and pushing changes into production automatically.

Distinctions

2024	Best Thesis Research Plan (TRP) Award by Doctoral Program in Electrical and Computer Engineering, Faculty of Engineering, University of Porto (FEUP) attributed to my PhD's TRP
2020	European Commission Innovation Radar by European Commission, Belgium attributed to FotoInMotion: Repurposing and enriching images for immersive video storytelling